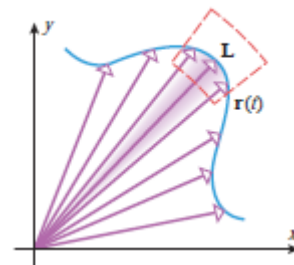


2.2 Calculus of Vector-Valued Functions

Recall: A vector-valued function $\mathbf{r}(t)$ maps real numbers to n -dimensional vectors $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$.

It is important to visualize how $\mathbf{r}(t)$ plots points via vectors.

We begin this section by looking at the limit of a vector-valued function.



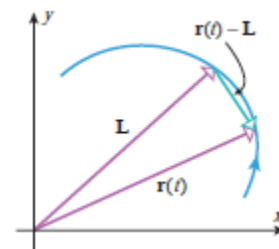
Definition: Let $\mathbf{r}(t)$ be a vector-valued function that is defined for all t in some open interval containing the number a (it need not be defined at a). We write that

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{L}$$

if and only if

$$\lim_{t \rightarrow a} \|\mathbf{r}(t) - \mathbf{L}\| = 0$$

Note: It is important to see that the limit of $\mathbf{r}(t)$ is a vector $\mathbf{L}$. $\mathbf{L}$ is often referred to as a limiting vector. Therefore, we don't imagine we following points on the curve when we evaluate the limit of $\mathbf{r}(t)$, but rather we are following the sequence of vectors since $\mathbf{r}(t)$ is a function that produces vectors.



The above definition suggests the following theorem.

Theorem: If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then we define $\lim_{t \rightarrow a} \mathbf{r}(t)$ as follows:

Proof: Omitted, but it follows from the above definition.

Definition: A vector function $\mathbf{r}(t)$ is continuous at a if

Theorem: A vector function $\mathbf{r}$ is continuous at a if and only if its component functions are continuous at a .

Proof: This theorem follows immediately from how continuity is defined for vector functions and how continuity is defined in calculus one for functions of the form $f: \mathbb{R} \rightarrow \mathbb{R}$.

Example 1: Consider the vector function $\mathbf{r}(t) = \left(\frac{t^2 - t}{t - 1} \right) \mathbf{i} + (\sqrt{t + 8}) \mathbf{j} + \left(\frac{\sin(\pi t)}{\ln(t)} \right) \mathbf{k}$.

a) Find $\lim_{t \rightarrow 1} \mathbf{r}(t)$

b) Is the function $\mathbf{r}$ continuous at 1? Give a brief explanation of your answer.

Definition: The derivative of a vector function $\mathbf{r}$ is defined by: $\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$.

Intuitively, one can imagine that as h approaches 0 the vector in orange will approach the vector in green (which is tangent to the curve at the terminal point of $\mathbf{r}(t)$).

This leads naturally to the following definitions.

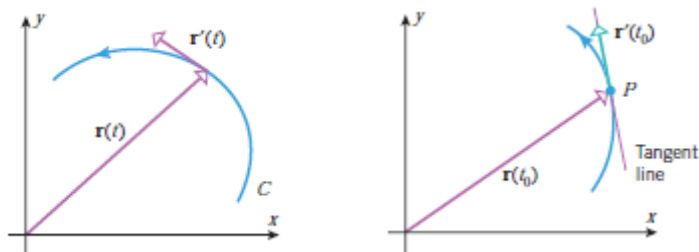
o*



Definition: Suppose that $\mathbf{r}'(t) \neq \mathbf{0}$. The vector $\mathbf{r}'(t)$ is called the tangent vector to the curve, C , defined by $\mathbf{r}$ at the terminal point, P , of the position vector $\mathbf{r}(t)$.

The tangent line to C at P is defined to be the line through P parallel to $\mathbf{r}'(t)$.

Also, $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ is called the unit tangent vector.



The next theorem gives us a familiar way to calculate $\mathbf{r}'(t)$.

Theorem: If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ and the component functions of $\mathbf{r}$ are differentiable then,
 $\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$.

Proof:

Note: There is nothing special in the above proof that requires the vector function to have 3 components. The same proof works for a vector function of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$.

Example 2: Consider $\mathbf{r}(t) = \langle 2 \cos(t), \sin(t), t \rangle$

a) Find $\mathbf{r}'(t)$

b) Find $\mathbf{T}(t)$

c) Find parametric equations for the tangent line to the curve determined by $\mathbf{r}$ at the point $\left(0, 1, \frac{\pi}{2}\right)$.

Note: You can use the graphing software on Blackboard to help visualize what we have found in this example.

Theorem: Suppose $\mathbf{u}: \mathbb{R} \rightarrow \mathbb{R}^3$ and $\mathbf{v}: \mathbb{R} \rightarrow \mathbb{R}^3$ are differentiable vector functions, c is a scalar, and $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function. Then,

1. $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
2. $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$
3. $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
4. $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
5. $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
6. $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

Proof: Many of these are left as practice in the exercises for this section.

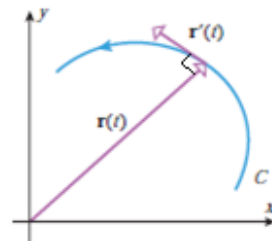
Note: With the exception of property 5 all of the above properties hold when $\mathbf{u}$ and $\mathbf{v}$ have two or more component functions.

Theorem: If $\mathbf{r}(t)$ is a differentiable vector-valued function in 2-space or 3-space and $\|\mathbf{r}(t)\|$ is constant for all t , then

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t .

Proof:



Definition: The definite integral of a continuous vector function $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ is

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle.$$

Note: For now we will hold off on a physical interpretation of the definite integral (we will see a nice application for it in 2.6). The indefinite integral is defined similarly.

Example 3: Suppose $\mathbf{r}(t) = (\cos(t))\mathbf{i} + \left(\frac{1}{1+t^2}\right)\mathbf{j} + (\sin^2(t))\mathbf{k}$. Find $\int_0^{\frac{\pi}{4}} \mathbf{r}(t) dt$ (exactly).

Note: Again, it is important to see that our result is a vector.

Example 4: Suppose $\mathbf{r}(t) = (\cos(t))\mathbf{i} + \left(\frac{1}{1+t^2}\right)\mathbf{j} + (\sin^2(t))\mathbf{k}$. Find $\int \mathbf{r}(t) dt$.

Note: If we had been given some initial condition regarding the vector-function $\int \mathbf{r}(t) dt$, then we could have solved for the values of all the unknown constants. This is just like initial condition questions from Calculus I.